

Bachelor Project

AI Driven Analysis of Kidneys in CT Scans

Weekly Report 5

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What has been done this week

1.1 Annotation

I have made the final annotations, so now I have a full test set of 10 CT scans and a corresponding mask for cortex and medulla in the left kidney.

1.2 Testing model proposed by Korfiatis et al. [1]

Korfiatis et al. have made their model for cortex-medulla segmentation and an inference script available on <https://github.com/potis/MedCorSeg>. Running the script does preprocessing as well (windowing, normalizing, etc.). They used five classes (background, left cortex, right cortex, left medulla, and right medulla). They also provided an example of a subvolume of a CT scan, cropped around the kidneys.

1.2.1 Example provided by authors

I had some trouble running their code, since they used some old versions of python and tensorflow, but I made it work on a computer I borrowed. First I ran inference on the example CT from the repository, the results can be seen in figure 1.

1.2.2 Cropping

I tried running the model on a couple of the whole-body CT scans from TotalSegmentator just to see if cropping was necessary, but the predictions were

very bad, so I decided to crop the images around the kidneys (since the model was trained to classify cortex and medulla in both kidneys, I thought it would be best to keep both kidneys in the image and not just the left one).

First, I wanted to crop to the same size as the example subvolume provided by the authors, but with the dimensions of the whole-body scans this was hard. Instead, I did the following:

For each image in the test set I used the corresponding whole-kidney masks for both kidneys (from TotalSegmentator), combined them to one mask, and used this to make a bounding box around the kidneys in the image. I added a margin of 15 voxels in each dimension and then extracted the box with margins from the image. As a result of this, the cropped images in the test set have different sizes.

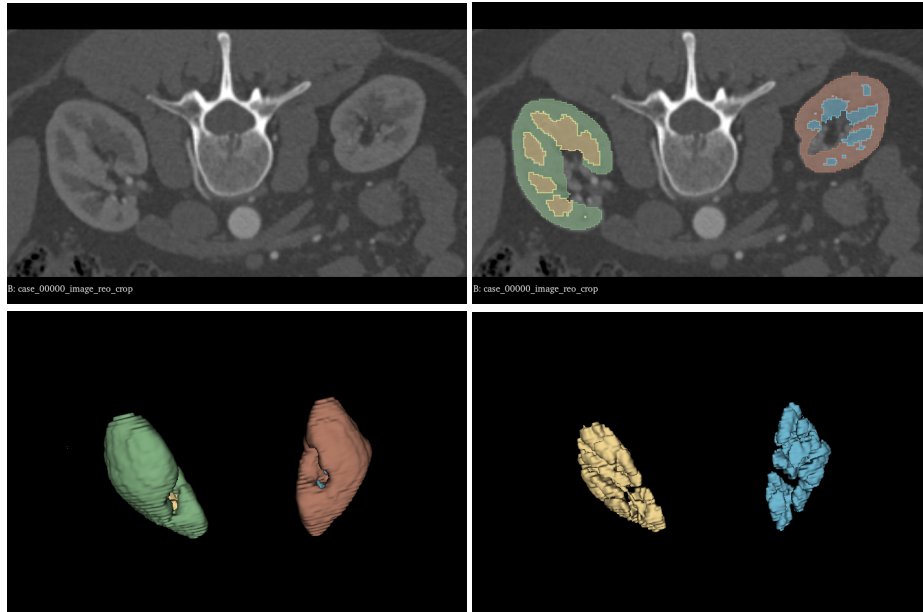


Figure 1: Predictions of the cortex-medulla segmentation model, proposed by Korfiatis et al. [1], on the example CT subvolume provided by the authors. Class colours are green (right cortex), yellow (right medulla), red (left cortex), and blue (left medulla). ***NB: I'm not 100% sure which are left and right.***

1.2.3 Predictions on test set

Running the model on the cropped images gave different results. for instance, the results were much better for subject s0065 (see figure 2) than for subject s1372 (see figure 3).

Generally, the model has a hard time differentiating between left and right in many slices. I don't know if this could have something to do with the fact that the images in the test set have the spine placed in the bottom part of the image when seen in axial view. The example CT from the repository has the spine

placed in the upper part of the image, so if the model is trained on images like that, then that might make a difference?

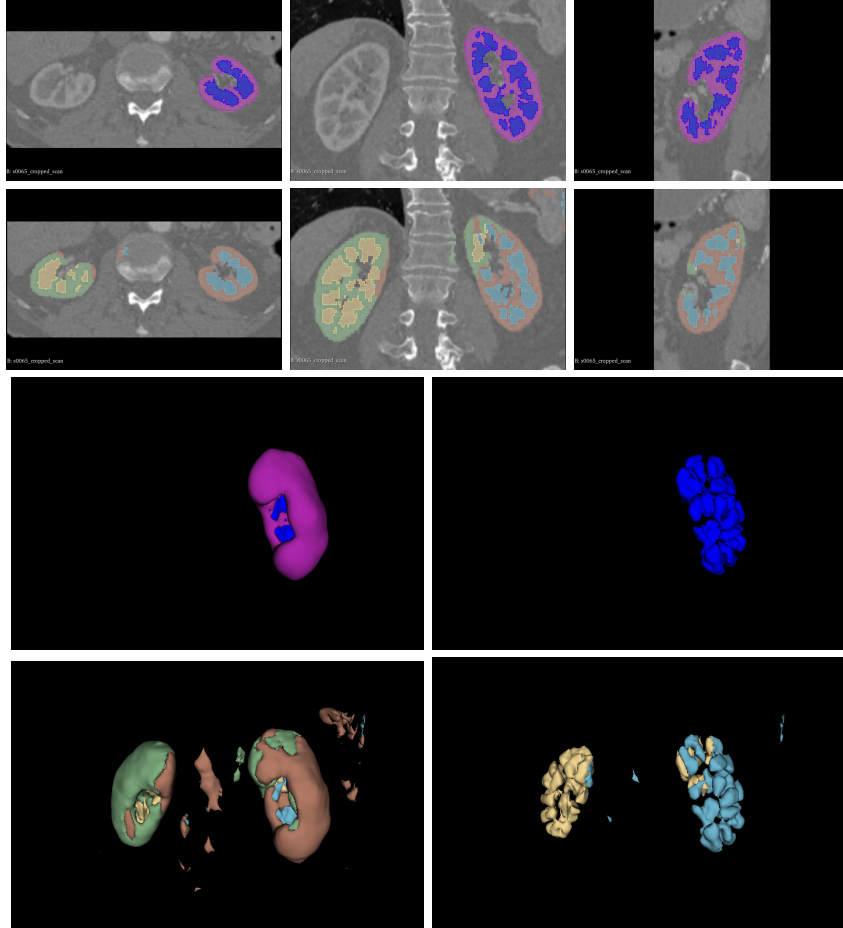


Figure 2: Ground truth vs. predictions of the cortex-medulla segmentation model, proposed by Korfiatis et al. [1], on subject s0065 in the test set. Predicted classes are coloured green (right cortex), yellow (right medulla), red (left cortex), and blue (left medulla). ***NB: I'm not 100% sure which predictions are left and right.*** Ground truth for left cortex is magenta and and ground truth for left medulla is dark blue.

(In general I'm confused about why the example and the test images look different regarding the spine in axial view since the small man (human coordinate) in Slicer looks the same both for the example image and the test images - see figure 4)

If we ignore the fact that the model classifies tissue in the left kidney as right and vice versa, it looks like it often gets cortex vs. medulla correct within the true masks. However, it also marks a lot of background voxels as cortex and medulla for some subjects.

When inspecting the test images it seems there is a bit of a difference between how long after contrast enhancement the scan is made, and generally there is a larger contrast between cortex and medulla in the example image from the repository than in the test images, which suggests that different CT phases might be used, see figure 4. Maybe this influences the ability to distinguish between cortex-medulla and the surrounding organs.

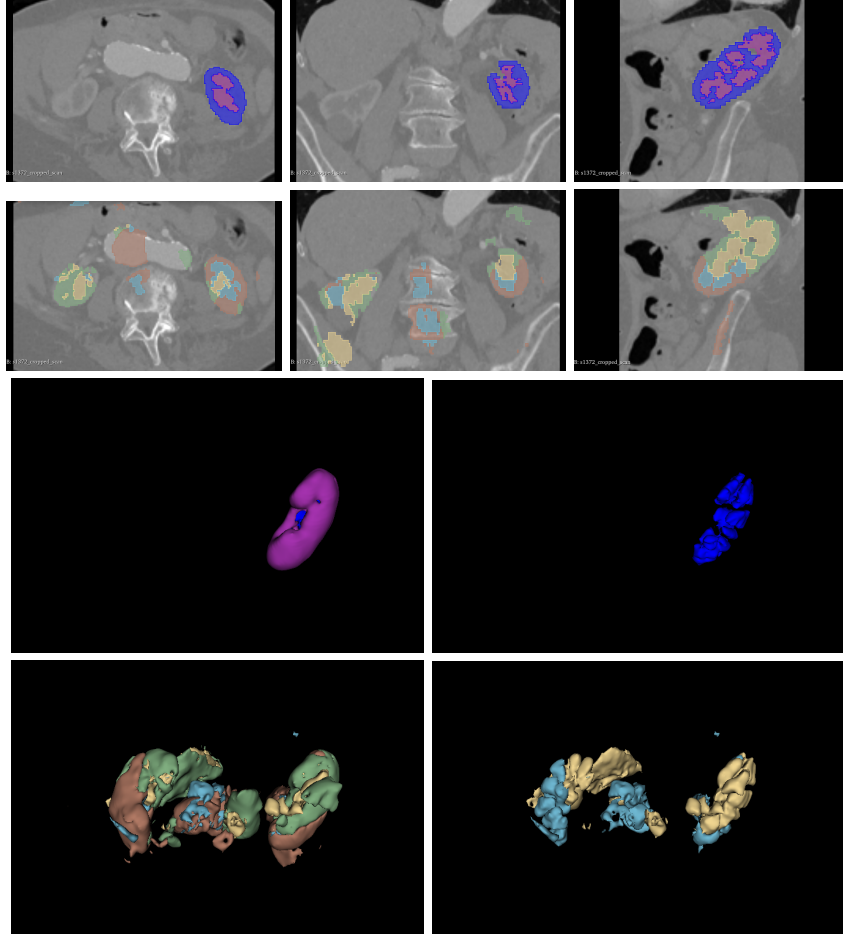


Figure 3: Ground truth vs. predictions of the cortex-medulla segmentation model, proposed by Korfiatis et al. [1], on subject s1372 in the test set. Predicted classes are coloured green (right cortex), yellow (right medulla), red (left cortex), and blue (left medulla). ***NB: I'm not 100% sure which predictions are left and right.*** Ground truth for left cortex is magenta and and ground truth for left medulla is dark blue.



Figure 4: Possible phase difference seen in three scans in axial view. The human coordinate system (small man in right corner) is oriented the same way for all three images, although the example image from the repository has the spine in the upper part, while the others have the spine in the bottom part.

Project status according to the study plan

It is going okay, but my initial project plan I made for myself ended around here, so I am not completely sure what the next steps are.

Question: In general when I implement a framework and only have a test set with annotations of size 10, is it then okay to look at the predictions on the test set and before making choices for postprocessing, although the test set should be independent? I am thinking whether I should in fact run the model on some of the images that are not in the test set and then look at the predictions, even though I don't have annotations, and take these into consideration when making choices about postprocessing? Then I would only use the test set in the end and it wouldn't be involved in any kind of development of the framework? If that makes sense? :)

Plan for the next weeks

- Find out what to do about left/right problems - should I maybe just not distinguish between left and right labels in the left kidney, but only cortex and medulla labels? I guess when we know in what side of the image the left kidney should be, then we can just cut the image in half and look at cortex/medulla labels in general in that half image?
- Do some postprocessing to remove small cortex-medulla parts that are not connected to the rest. I don't think this will fix the problem for the cases where a lot of background is labeled as cortex or medulla though.
- When the above is done, then I should compute metrics (DSC, accuracy etc.)
- Try another framework? Or use this framework on a larger dataset (preferably with phase similar to the example image) and see if I can get results and compute cortex-medulla volume from this?

References

- [1] Panagiotis Korfiatis, Aleksandar Denic, Marie E. Edwards, Adriana V. Gregory, Darryl E. Wright, Aidan Mullan, Joshua Augustine, Andrew D. Rule, and Timothy L. Kline. Automated segmentation of kidney cortex and medulla in CT images: A multisite evaluation study. *Journal of the American Society of Nephrology*, 33(2):420–430, February 2022.